

COMP 106: DISCRETE MATHEMATICS FOR COMPUTER SCIENCE AND ENGINEERING (Fall 2024 Syllabus)

Instructor: Alptekin Küpçü

TAs: Check course web page.

Office Hours: Check course calendar.

Webpage: <https://courses.ku.edu.tr/comp106/>

E-mail: Always send email to `comp106ta@ku` for course-related questions. **All other emails will be ignored!**

Topics:

- Logic, proof techniques, sets, functions.
- Relations, equivalence classes.
- Algorithms, complexity of algorithms.
- Mathematical induction, recursive algorithms.
- Theory of computation, languages, grammars, finite state machines, Turing machines, complexity classes.

Objectives:

1. To teach about fundamentals of logic and mathematical thinking, functions, predicates, sets, relations, formal proof techniques.
2. To introduce you to mathematical induction and recursion.
3. To provide a basic understanding of algorithms, complexity, and theory of computation.
4. To prepare you for success in the following years.

Learning Outcomes: At the end of this course, you will be able to

1. form proper logical statements and mathematical proofs,
2. analyze time complexity of basic algorithms, including recursive algorithms,
3. grasp basic concepts of theory of computation, such as languages, grammars, finite state machines, and Turing machines.

Text Book: Kenneth H. Rosen, “Discrete Mathematics and Its Applications”, 6th or 8th Edition, ISBN: 978-0073229720. (Required)
Susanna S. Epp, “Discrete Mathematics with Applications”, 4th Edition, ISBN: 978-0495391326. (Recommended)

Grading: Quizzes (40%), Midterm Exam (30%), Final Exam (30%).

- You have to obtain at least 40/100 out of each exam to pass the course.
- Catalog-like system, no curve grading. This means, others’ performance does not affect your grade.
- Grading objection must be done within 3 days following the grade announcement.
- If you have an approved health report for midterm exam, your final exam grade will also be counted as your midterm exam grade.

- If you have an approved health report for the final exam, please apply for the remedial exam.
- You have to request makeup by indicating your approved health report.

Code of Conduct:

Attendance: Please do not step in and out of the classroom when the lecture/PS is in progress (in important cases, ask for my permission *ahead of time*). TAs have the same authority as the instructor, and hence you must follow their rules as well.

Device Use: Use of devices such as laptops, cell phones, or similar devices is **strictly forbidden** in the classroom.

Assignments: **Quizzes will be during PS hours.** Hand-written solutions will be accepted as long as they are legible. It is our right to cut off grade points if we cannot read your answer. Always target *the best possible asymptotic complexity* for full grade.

Mandatory: For each assignment, you must sign the following (otherwise your assignment will not be graded): “I hereby declare that I have completed this individually, without support from anyone else. I hereby accept that only the below listed sources are approved to be used:

- (i) Course textbook,
- (ii) All material that is made available to me via Blackboard for this course,
- (iii) Notes taken by me during lectures.

I have not used, accessed or taken any unpermitted information from any other source. Hence, all effort belongs to me.”

Name Surname, Date, Signature (all written with blue pen).

Cheating: Do not cheat. We employ **cheating detection tools** that can easily detect cheating. By national and university-wide rules, any cheating *attempt* may be penalized with **one semester suspension** from the university. The Koç University Student Code of Conduct includes, but is not limited to, the following: Your work needs to be your own. It cannot be joint work with any other person (in the class or not). You cannot have someone do the assignment for you (paid or not). You cannot submit an assignment that is based on submissions in earlier semesters. You cannot use a solution manual, or an online solution to the question. To get a grade (other than F) in this course, you must fully comply with the Koç University Student Code of Conduct as seen on:
<https://apdd.ku.edu.tr/en/academic-policies/student-code-of-conduct-2/>
 Attending any Koç University course means you have read and agreed to this conduct, the course syllabus, and honor codes. Failure to comply will result in a **direct F for your letter grade for the course and disciplinary action.**

- How to Succeed:**
1. You need to do **lots of reading**. Do not expect to understand hard topics without reading from *multiple* sources.
 2. Make sure to do the assignments and suggested exercises yourself, without finding the solution somewhere else.
 3. You need to solve **more exercises than the ones assigned**. The assignments will help, but they will **not** be enough to guarantee success.
 4. If you did not understand a particular topic, **ask** before we switch to the next topic.
 5. **Work regularly**. It is nearly impossible to succeed in this course by just studying for exams at the last minute.

6. Failure to notice an announcement is not an excuse. It is your responsibility to keep up-to-date with the course material.
7. Do not forget that this is one of the most fundamental and important courses for Computer Engineering. Try to learn the topics in the best possible way.
8. **Do not cheat.**